

ChiPi Technologies

Industry-Guided Project Presentation

Project Experience Initiative • Real Projects • Real Mentorship • Real Outcomes

Student Profile

Name:	Raja Babu
College:	Impact Group of Colleges, Patna
Branch:	Bachelor of Computer Applications (BCA)
Batch:	2024 - Present

Project Overview

Executive Summary

Engineered and deployed a multi-page personal portfolio and professional resume web application consisting of three interconnected modules (`index.html` , `My_learning.html` , and `My_Project.html`). The system acts as a centralized digital profile displaying core technical qualifications, an active education timeline, live source code repositories, and an interactive project catalog optimized for professional networking and career presentation.

Problem Statement

The Challenge Being Solved

- Aspiring developers struggle to showcase a multi-faceted background (encompassing academic studies, prior multimedia design tools, and real-time learning tracks) inside a static paper resume.
- Traditional single-page resumes lack interactive capabilities to seamlessly bridge professional summaries with live code repositories and sub-project indexes.
- Need for an interconnected, highly accessible digital portal to establish technical credibility during early-stage career recruitment.

Business Impact

Who Benefits

- **Recruiters & Managers:** Gain immediate, friction-free access to a candidate's complete profile and active GitHub repositories.
- **The Candidate:** Accelerates placement opportunities by presenting proof of self-directed technical executions.

How They Benefit

- Reduces screening time via centralized documentation layout.
- Provides live technical validation of frontend layout principles and structural engineering.

Requirements

Functional Requirements

- Modular, multi-page layout traversing profile bios, active learning sheets, and standalone code indices.
- Persistent global navigation headers to enable frictionless relative path traversal.
- External hyperlink integration pointing securely to exterior validation nodes (GitHub / LinkedIn).

Non-Functional Requirements

- Strict content isolation and decoupling using external style rules (CSS3).
- Optimized legibility utilizing clean, professional typography families (Arial/ Sans-Serif).
- High contrast container frames and distinct structural borders for intuitive reading layout.

Solution Architecture

System Design

The application follows a clean decoupled presentation pattern where individual views inherit shared layout properties from a centralized styling master layer:

style.css (Central Layout & Visual Schemes)

index.html

Profile & Summary

My_learning.html

Skills Matrix Track

My_Project.html

Portfolio Index

UI/UX Design

Layout Aesthetics & Wireframe Definition

- **Asymmetrical Layout:** Engineered a dual-panel visual presentation on the core index page using specialized column grouping to segment high-impact text.
- **Left Accent Panel:** Configured with a deep, professional linear gradient backdrop mapping from `#2f2f2f` down to `#886868` to frame critical quick-scan items (photo, contact links, core skills).
- **Right Canvas Panel:** Styled with an elegant light tone background (`rgb(232, 222, 222)`) ensuring perfect text readability, high color contrast ratios, and comfortable skimming of academic achievements.

Technology Stack

Tools & Technologies Utilized

Core Frontend Architecture:

HTML5

CSS3 Blueprinting

Linear Gradients

Box Shadow Layering

Development Ecosystem & Workflows:

Git Version Controls

GitHub Code Hubs

Asymmetric Wireframing

Academic Framework Context:

C / C++ Foundations

Java Core

MySQL Database Management

Rust Language Track

Implementation Journey

Development Process Phases

1. **Phase 1 - Structural Blueprinting:** Documented core biographical assets and established localized tables to frame out multi-panel architectures.
2. **Phase 2 - Style Decoupling:** Transferred baseline container styles to an autonomous layout file (`style.css`), establishing universal definitions for cards, shadows, and list attributes.
3. **Phase 3 - Link Orchestration:** Built relative internal links across all three documents and embedded external targeting hooks to connect active online profiles.

Challenges Faced

Obstacles Encountered

- **Component Layout Stability:** Maintaining symmetrical padding structures and preventing text overlapping when styling multi-panel displays across varying initial view sizes.
- **Navigation Theme Cohesion:** Preventing sudden structural shifts or tracking discrepancies when routing the user across independent sub-documents.
- **Asset Integration Control:** Isolating profile image dimensions gracefully within dynamic layout panels using local relative references.

Solutions Applied

How Challenges Were Resolved

- **Explicit Panel Segmentation:** Implemented explicit dimension attributes and forced alignment bindings (`valign="top"`) to guarantee structural containment across separate content areas.
- **Global Style Inheritance:** Linked a standardized external style script across all sub-pages, enforcing global font rendering behaviors (`Arial, sans-serif`) and list aesthetics universally.
- **Object-Fit Constraint Styling:** Managed profile photos utilizing modern CSS formatting keys (`object-fit: cover` and `border-radius: 8%`) to lock image aspect ratios.

Code Walkthrough

Multi-Page Link Navigation:

```
<table width="100%" bgcolor="Gray" height="40">
  <tr>
    <td align="left" width="45%">
      <a href="./index.html"><b>About_me</b></a>
    </td>
    <td width="40%">
      <a href="./My_learning.html"><b>My learning</b></a>
    </td>
  </tr>
</table>
```

Isolated Global Shadows & Formats:

```
ul {
  width: 80%;
  margin: 20px auto;
  padding: 20px 40px;
  border-radius: 10px;
  box-shadow: 0 0 10px rgba(0,0,0,0.1);
}

ul li {
  margin: 10px 0;
  font-size: 18px;
}
```

Testing & Validation

Testing Evidence & Verification Steps

- **Relative Path Sanity Checks:** Performed localized functional walkthroughs to verify that navigating between `About_me` , `My learning` , and `My project` elements remains fully operational without generating broken links or 404 response codes.
- **External Anchor Validations:** Manually audited outgoing link schemas on `My_learning.html` to verify targeted external loops trigger direct context routing into specified user endpoints on GitHub and LinkedIn.
- **CSS Style Rendering Audits:** Assured custom card rounding metrics (`border-radius` rules) and drop shadows render consistently across layout updates.

Results & Achievements

Core Artifacts Deployed

- **3 Standalone Modules:** Built an interconnected digital profile ecosystem with flawless navigation continuity.
- **Zero Inline Layout Style Overhead:** Clean separation of style and structure for easier code maintenance.

Engineering Milestones

- Successful integration of live portfolio linkages.
- Robust layout control via centralized class rules and box constraints.

Key Learning

Technical & Professional Growth

- **Architectural Mastery:** Acquired hands-on proficiency executing decoupled frontend designs, managing local relative layout states, and formulating targeted style overrides.
- **Strategic Digital Presentation:** Developed a professional portfolio structure that maps out immediate academic targets (BCA Semester 4 tracking) alongside highly demanding programming ecosystems (Rust and MySQL tracks).
- **Career Acceleration:** Formulated an engineering-first resume strategy by framing typical educational profiles into high-visibility software portfolios.